

Fire Extinguishing Devices Trade-Off

Technology Suggestions	Red Powder/ Dry Chemical ("Fire Retardants")	Extinguishing Grenade/ Dry Powder Bomb	Water
Criteria			
Area Efficiency per mass	Covers a high area per kg 4.21 m ² per kg [1]	30 kg dry powder bomb diffuses to blanket an area between 130 and 700 square meters: 4.3 to 23.3 m ² per kg [2]	Poor coverage 0.01 to 0.2 m ² per kg [3]
Area Efficiency per flight minute	~350 to 400 m ² per minute [4]	~30 m ² per minute [2]	Heavy mass limits discharge time ~210 to 310 m ² per minute [5]
Precision Tolerance	Does not have to be precise as broad cloud covers broad areas	Precise kinetic delivery, does not succumb so easily to wind but the trajectory upon impact must be precise to target fire correctly. can cover much area if exploding takes place a few meters before hitting, one-shot impact, high localized area coverage, and does not depend too much on wind conditions to reach target. Better control on where it falls compared to other solutions.	Requires direct continuous hits
Total Extinguishing and risk of reignition	Rapid chemical knockdown, creates long-term barriers Poor thermal cooling => high risk of reignition	Extremely effective on knocking canopy flames. The kinetic blast of a dry powder bomb knocks out active flames instantly by starving them of oxygen. However, it does nothing to lower the core temperature of thick wood => very high risk of reignition	Prevents reignition by thermally cooling fuel (when not evaporating too much...)
Suitability for outdoors (wind)	Wind assists the overall distribution which is good	Yes. Wind changes its trajectory the least.	Yes
Environmental impact	Ammonium polyphosphates: Toxic, especially for aquatic ecosystems (fish) and vegetation	ABC Monoammonium Phosphate: Very toxic, especially for aquatic ecosystems (fish), vegetation and terrestrial wildlife	No
Cost	Moderate	High	Low
Availability/ Accessibility	Restricted (Requires advanced manufacturing and mixing bases)	Highly Accessible (Stored easily in local fire stations)	Theoretically infinite and immediate (but depends on location and up to water availability)

Weighing factors legend:

Importance of criteria: 1-7

Marks: 1-3

Technology Suggestions	Red Powder/ Dry Chemical ("Fire Retardants")	Extinguishing Grenade/ Dry Powder Bomb	Water
Criteria			
Area Efficiency per mass (7)	2	3	1
Area Efficiency per flight minute (6)	3	1	2
Precision Tolerance (3)	2	3	1
Total Extinguishing and risk of reignition (3)	1	2	3
Suitability for outdoors (Type A Fires) (5)	3	3	3
Environmental impact (2)	1	1	3
Cost (1)	2	1	3
Availability/ Accessibility (2)	1	2	3
TOTAL:	62	64	61

Red Powder Examples:

- <https://www.perimeter-solutions.com/product/phos-chek-259-fx-2/>
- <https://xpyro.fi/> (more eco-friendly)

Dry Powder Bomb Examples:

- <https://jinhongdrone.com/fire-extinguishing-bomb-deployment-device> (bombs (23 kg each) with 20 kg extinguishing agent per unit, providing coverage of $\geq 50 \text{ m}^2$ and safe wireless deployment)
- <https://www.youtube.com/shorts/bTtY0pKd4pU>
- How it works: <https://salgrom.fi/en/how-does-impulse-dry-powder-fire-suppression-work/>

To be noted: Water deployment remains indispensable at some stage of suppression because it absorbs residual thermal energy to prevent reignition. While these tactics complement one another, payload-limited UAVs should prioritize precision chemical retardants or dry-powder bomb strategies rather than bulk water delivery, which requires a much heavier cargo capacity.

References

[1]

https://www.google.com/url?sa=i&source=web&rct=j&url=/goto?url%3DCAESlgEB6zswFaW6jig_nAzX5a_wuSaun9QoBNS2FJNdq4DhdCqCsjF2YeGIPwS30PeL7lgwNpzXbBMquxrUHsJ1aPgX7LqV9FkTJmAQVq-u6UTnTqVSLL_fguXw6LhQijW65jju9XBnDlppdOvEpJsm8bgwboF16mUq8fn1g9E8_LpZjDJF-7ScNL422HSru0BzK4vkxFvbdXI&ved=2ahUKEwiqssbbkiOXAxVdHhAIHW0QHMQQqYcPegYIAAgnEAI&opi=89978449&cd&psig=AOvVaw0Knjlk9wtdBAUFHfSz9hOf&ust=1790198896298000

[2]

<https://jinghongdrone.com/uav-for-fire-fighting-key-technologies-and-features-explained>

[3]

<https://wmpllc.org/ojs/index.php/jem/article/view/2689>

[4]

https://www.fs.usda.gov/t-d/pubs/pdfpubs/pdf00572851/pdf00572851_72dpi.pdf

[5]

https://en.wikipedia.org/wiki/Canadair_CL-415